先进计算与数字工程研究中心

PATTERN RECOGNITION AND MACHINE LEARNING

Lecture 05 – Discriminative Classifiers II: Perceptrons and Decision Trees



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A spam filter learns from a wrong prediction

- A message is marked as safe, but the user reports spam.
- The perceptron changes its weights so this message receives a higher spam score.
- A decision tree can instead ask successive questions about the message.

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Syllabus boundary for Lecture 05

Syllabus topic	Allocated time	Class emphasis
Perceptron algorithm	1.0 hour	updates, convergence, failure
Decision trees	1.5 hours	partition, split, pruning
Integrated practice	0.5 hour	coding and case reasoning

Deliberate boundary

Feature engineering begins in Lecture 06. Random forests wait until the ensemble lectures.

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Linear scores and decision trees

- A linear classifier computes one weighted score for the whole input space.
- A tree asks one feature question at each internal node.
- Several tree branches can represent a boundary that one line cannot.

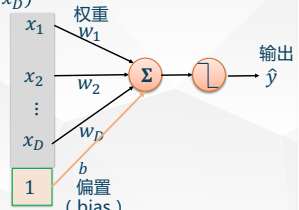
The perceptron computes a weighted score

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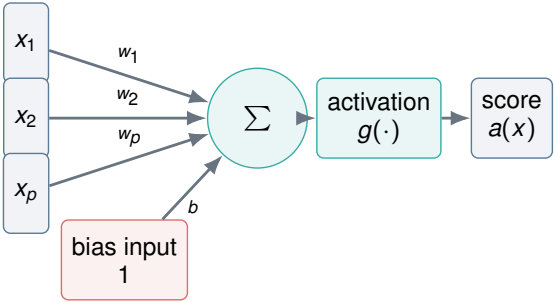
感知器（ Perception ）算法

- 感知器是由美国学者Frank Rosenblatt在1957年提出来的，是神经网络和支持向量机的基础。
- 感知器是一种二类分类的线性分类模型
  - 输入为实例的特征向量$\boldsymbol{x} = (x_1, x_2, \dots, x_D)^{\mathrm{T}}$
  - 输出为实例的类别：
    $y = +1$代表正类
    $y = -1$代表负类
- 线性判别函数：$f(\boldsymbol{x}) = \boldsymbol{x}^{\mathrm{T}}\boldsymbol{w} + b$
- 判断规则：

$$\hat{y} = \mathrm{sgn}\big(f(\boldsymbol{x})\big) = \begin{cases} 1 & f(\boldsymbol{x}) > 0 \\ -1 & f(\boldsymbol{x}) \le 0 \end{cases}$$



Bias inside an augmented vector



Augmented sample

$\tilde{x} = [x^\top, 1]^\top$

Augmented weight

$\tilde{w} = [w^\top, b]^\top$

$$a(x) = \tilde{w}^\top \tilde{x}$$

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Labels become signs

Convention

Use $y_i \in \{-1, +1\}$.

$$y_i \, \tilde{\boldsymbol{w}}^\top \tilde{\boldsymbol{x}}_i > 0$$

One test

A correct prediction makes the signed score positive.

The update reacts only to a mistake

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Correct sample

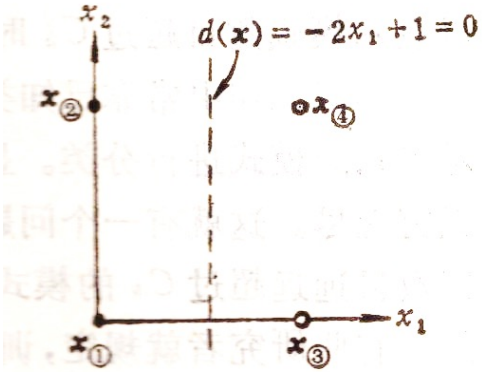
$y_i \tilde{\boldsymbol{w}}^\top \tilde{\boldsymbol{x}}_i > 0$ gives no change.

Mistake

$y_i \tilde{\boldsymbol{w}}^\top \tilde{\boldsymbol{x}}_i \leq 0$

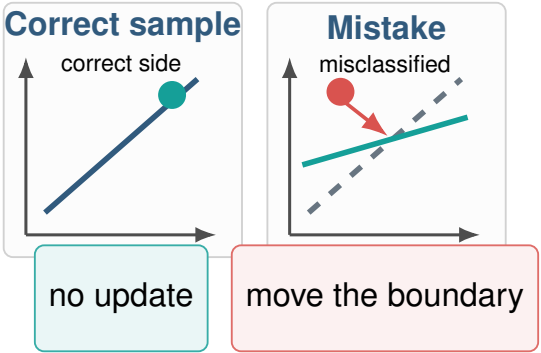
$$\tilde{\boldsymbol{w}} \leftarrow \tilde{\boldsymbol{w}} + \eta y_i \tilde{\boldsymbol{x}}_i$$

Follow a perceptron update



The current score is $-2x_1 + 1$; a mistake changes both weights and bias.

Reward and punishment as an early metaphor



Historical wording

Correct patterns receive no penalty. Misclassified patterns push the weight vector.

Modern reading

The rule optimizes a classification criterion. It does not imitate a biological neuron in detail.

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A four point training set

Sample	$\tilde{x} = (x_1, x_2, 1)$	y
A	(0,0,1)	+1
B	(0,1,1)	+1
C	(1,0,1)	-1
D	(1,1,1)	-1

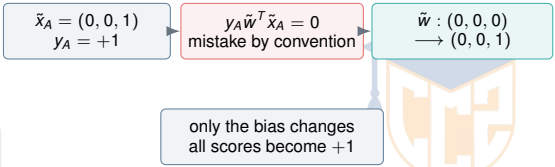
Initial state

$\tilde{w}^{(0)} = (0, 0, 0)$ and $\eta = 1$.

The first mistake creates a bias

$$y_A \tilde{w}^{(0)\top} \tilde{x}_A = 0$$

$$\tilde{w}^{(1)} = (0,0,0) + (0,0,1) = (0,0,1)$$



Meaning

The classifier now predicts every point as positive.

A negative mistake changes the slope

$$y_C \tilde{w}^\top \tilde{x}_C = (-1)(1) = -1$$

$$\tilde{w} \leftarrow (0,0,1) - (1,0,1) = (-1,0,0)$$

Boundary

$-x_1 = 0$

Several epochs can be necessary

Epoch

One pass through all training samples.

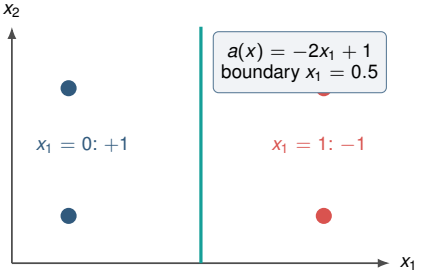
Stopping test

Finish an entire epoch with zero mistakes.

Bad shortcut

Stopping after a lucky local run can leave earlier samples misclassified.

The learned boundary in the classroom example



The second feature receives zero weight because it is irrelevant in this four-point example.

One valid solution

$\tilde{w} = (-2, 0, 1)$

$$a(x) = -2x_1 + 1$$

Check

$x_1 = 0$ gives $+1$ and $x_1 = 1$ gives -1.

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Perceptron learning loop

```
import numpy as np
X = np.array([[0.,0.,1.],[0.,1.,1.],[1.,0.,1.],[1.,1.,1.]])
y = np.array([1., 1., -1., -1.])
w = np.zeros(3)

for epoch in range(20):
    mistakes = 0
    for xi, yi in zip(X, y):
        if yi * (w @ xi) <= 0:
            w += yi * xi
            mistakes += 1
    print(epoch, mistakes, w)
    if mistakes == 0:
        break
```

Expected pattern

Mistakes decrease to zero for this separable data.

Learning rate changes the path

Update size

η controls how far one mistake moves the weight vector.

Separable case

A fixed positive learning rate preserves the convergence guarantee.

Finite budget

With early stopping, different learning rates can leave different classifiers.

Sample order changes the solution

Same constraints

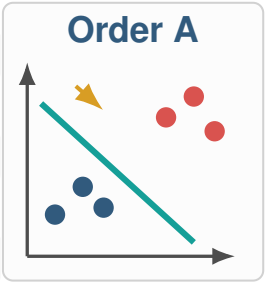
Every training point demands a positive signed score.

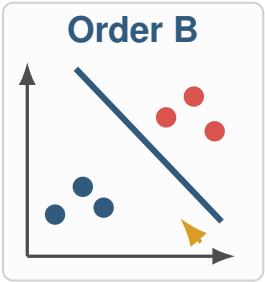
Different path

The next violated constraint depends on sample order.

Consequence

Two valid runs can end at different separating hyperplanes.





Convergence needs linear separability

Guarantee

For linearly separable data, repeated mistake updates reach a separating weight vector in finitely many mistakes.

Condition

The guarantee disappears when no separating hyperplane exists.

Margin explains an informal mistake bound

Data radius

Assume $\|x_i\| \leq R$.

Separating margin

Assume a unit vector u with $y_i u^\top x_i \geq \gamma > 0$.

$$M \leq (R/\gamma)^2$$

XOR cannot be separated by one line

- $y = 1$ for $(0, 1)$ and $(1, 0)$; $y = 0$ for $(0, 0)$ and $(1, 1)$.
- The two classes occupy opposite corners of a square.
- No single straight line separates these pairs of corners.
- Repeated perceptron updates cannot fix an impossible linear model.

Cycling is a diagnostic signal

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Observed pattern

Training mistakes repeat without reaching zero.

1. Check label errors and duplicate rows with conflicting labels.
2. Plot low dimensional projections.
3. Try a richer representation or a different model.



Do not claim convergence

A small recent mistake count is not a separating solution.

The pocket rule keeps the best visit

Idea

During a nonconvergent run, remember the weight vector with the fewest training mistakes.

Benefit

The final epoch no longer automatically decides the returned classifier.

Limit

Training error still needs validation on unseen data.

Different update orders give different boundaries

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思考

- 感知器算法存在许多解？
  - 初值的选择
  - 迭代过程中误分类点的选择顺序

Multiclass perceptron uses one score per class

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Scores

$a_k(x) = w_k^\top x$

$$\hat{y} = \arg\max_k a_k(x)$$

Interpretation

Each class owns a direction that competes for the sample.

A multiclass mistake updates two rows

If $\hat{y} \neq y$

$w_y \leftarrow w_y + \eta x$

Competing class

$w_{\hat{y}} \leftarrow w_{\hat{y}} - \eta x$

Effect

Raise the true class score and lower the mistaken winner.

One versus rest is a different construction

Construction	Training	Possible issue
Multiclass perceptron	joint competing scores	relative margins
One versus rest	independent binary models	scores may not align



Do not merge the names

Both use linear scores, but their training objectives differ.

Feature scale can dominate updates

Two features

Age ranges from 18 to 70 while annual spending ranges from 0 to 100000.

Update

The change $\eta y x$ inherits the units and scale of every feature.

Consequence

A large numeric range can dominate the path without being more informative.

Bias must follow the same update convention

Separate form

$w \leftarrow w + \eta y x$ and $b \leftarrow b + \eta y$

Augmented form

$\tilde{w} \leftarrow \tilde{w} + \eta y \tilde{x}$

Common bug

Updating w but forgetting b forces the boundary through the origin.

Prediction at exactly zero needs one rule

Option A

Treat $a(x) = 0$ as a mistake during training.

Option B

Break the prediction tie with a fixed class convention.

Requirement

Use the same convention in hand calculations, code, and evaluation.

Perceptron laboratory with a deliberate bug

```
def train_perceptron(X, y, epochs=30, eta=1.0):
    w = np.zeros(X.shape[1])
    for epoch in range(epochs):
        mistakes = 0
        for xi, yi in zip(X, y):
            if yi * (w @ xi) <= 0:
                w = w + eta * yi * xi
                mistakes += 1
        if mistakes == 0:
            return w, epoch + 1
    return w, epochs
```

Mutation

Replace <= with < and test a zero initialization.

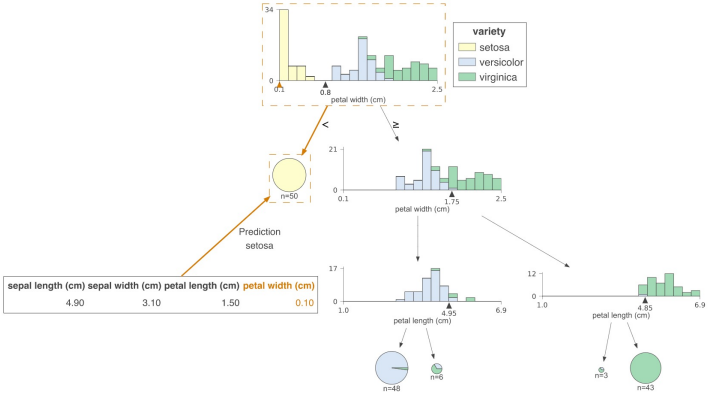
When the perceptron stops learning

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感知器算法的问题

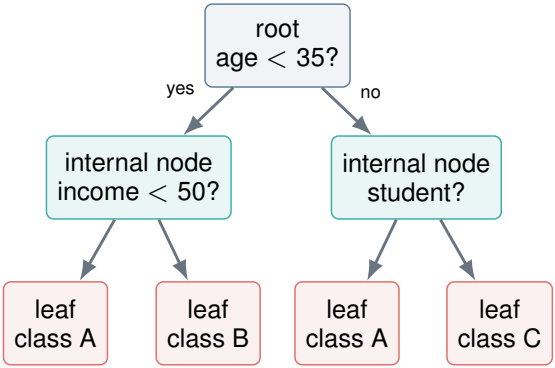
- 感知器算法只是当被分模式可用一个特定的判别界面分开时才收敛。在不可分情况下，训练过程始终不停止。
- 即使在模式可分的情况下，也很难事先算出达到收敛时所需要的迭代次数。
- 为此需要知道：发生迟迟不见收敛的情况时，到底是由于收敛速度过慢造成的，还是由于所给的训练样本集不是线性可分造成的？
- 最小平方误差(Least Mean Squared Error，LMSE)算法，对可分模式是收敛的，同时可指示类别不可分的情况。

Read a decision tree from root to leaf



Follow the feature test at each node to reach the prediction at a leaf.

Root, internal nodes, branches, and leaves



Root

The first test sees every training sample.

Internal node

A feature condition routes a subset.

Leaf

A prediction summarizes the samples that arrive there.

A branch records a test outcome; a leaf returns the prediction.

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A path is a conjunction of tests

$$R_m = \{x : x_{j_1} \le t_1,\ x_{j_2} > t_2, \ldots\}$$

Spoken rule

Low oxygen and high breathing rate lead to urgent review.

Logical detail

Conditions on different branches do not all apply to one sample.

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Recursive binary splits

分类回归树（Classification And Regression Tree，CART）

- ID3和C4.5根据特征取值，生成的树为多叉树
- CART：二分递归划分：将当前样本集合划分为两个子集为左右两个子节点，使得生成的每个非叶子结点都有两个分支 → 二叉树
  - 离散型特征：左右分支的组合较多（亦可将离散型特征编码成连续型特征）
  - 连续型特征：设特征X在$\mathcal{D}$中出现了M个不同的取值，将这些值从小到大排序，记作$a_1, a_2, \ldots, a_M$，则共有$M-1$个候选划分点依次为：$\frac{a_1+a_2}{2}, \frac{a_2+a_3}{2}, \ldots, \frac{a_{M-1}+a_M}{2}$。
  - 对大数据集，训练数据中特征的取值数目M可能会很多，要考虑所有$M-1$个候选点开销太大，可考虑将特征分成多个区间，采用直方图方式快速寻找近似的最佳划分点（XGBoost、LightGBM）

A classification leaf stores a class distribution

$$\hat{p}_{mk} = \frac{n_{mk}}{n_m}$$

$$\hat{y}_m = \arg\max_k \hat{p}_{mk}$$

Small leaf

A pure leaf with two samples gives a fragile probability estimate.

A regression leaf predicts a local constant

$$\hat{c}_m = \frac{1}{n_m} \sum_{i: x_i \in R_m} y_i$$

Model

$\hat{f}(x) = \sum_m \hat{c}_m \mathbb{I}(x \in R_m)$

Shape

The prediction changes in steps when a sample crosses a split.

A tree splits one feature at a time

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例：连续型特征

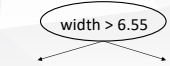
根据特征height的取值，对所有样本进行排序
得到可能的阈值为：
6.3，8.05，9.55

width	height	Class
6.9	6.0	Orange
6.0	6.6	Lemon
6.5	6.6	Lemon
6.6	6.6	Orange
7.0	9.5	Orange
6.9	9.6	Lemon

对每个可能的阈值，计算划分后的Gini指数：

$$H_{\text{heigh}>9.55}(\mathcal{D}) = H_{\text{heigh}>6.3}(\mathcal{D}) = \frac{1}{6}\times(0) + \frac{5}{6}\times\left(\frac{2}{5}\times\frac{3}{5}\times 2\right) = \frac{2}{5}$$

$$H_{\text{heigh}>8.05}(\mathcal{D}) = \frac{4}{6}\times\left(\frac{2}{4}\times\frac{2}{4}\times 2\right) + \frac{2}{6}\times\left(\frac{1}{2}\times\frac{1}{2}\times 2\right) = \frac{1}{2}$$



最佳决策规则为width > 6.55，划分后的Gini指数为$\frac{1}{4}$

Categorical questions need valid partitions

Nominal feature

Payment method has card, cash, transfer, and voucher values.

Valid split

Send a subset of categories left and the remaining categories right.

Invalid shortcut

Integer codes do not create a meaningful order for nominal categories.

Missing values require an explicit routing rule

Option

Impute before fitting the tree.

Option

Use a model that learns a default branch for missing values.

Audit

Missingness may itself carry information and may change after deployment.

Multiclass decisions fit naturally in leaves

Internal tests

The same binary questions can separate several classes step by step.

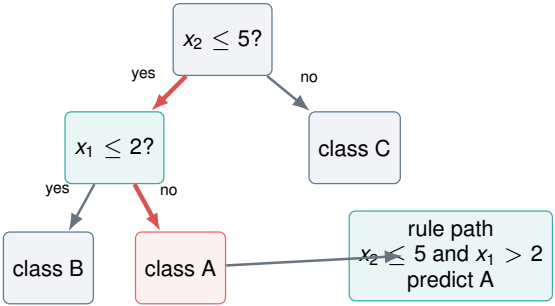
Leaf distribution

Each leaf may contain counts for every class.

Use case

A fault diagnosis tree can end in normal, sensor fault, motor fault, or overload.

A borrowed tree example becomes a rule list



Read the diagram

Start at the top test, follow one branch result, and stop at a leaf label.

Translation

If $x_2 \leq 5$ and then $x_1 > 2$, predict the leaf class shown on that path.

A tree explanation is conditional

Useful statement

This application reached a high risk leaf because debt ratio exceeded 0.55 after income fell below the first threshold.

Overclaim

The path does not prove that changing debt ratio will cause the outcome to change in reality.

Tree rules translate directly into code

```
def manual_tree(oxygen, breathing_rate):
    if oxygen < 92:
        return "urgent review"
    if breathing_rate > 24:
        return "observe"
    return "routine advice"
```

Trace

oxygen=95 and breathing_rate=27 reaches the second return.

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Depth controls rule complexity

Depth	Typical behavior	Risk
1	one stump	underfitting
3 to 5	short interactions	often readable
very deep	small specialized leaves	high variance

No universal best depth

Validation data must judge the tradeoff.

Deep trees have low bias and high variance

Low bias

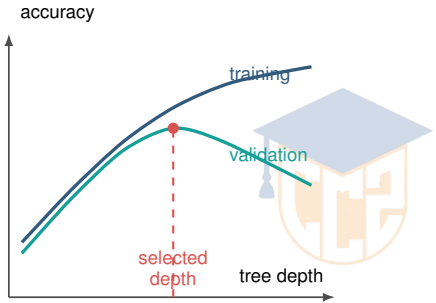
A sufficiently deep tree can fit intricate training patterns.

High variance

Small data changes can alter early splits and the entire subtree below them.

Symptom

Training accuracy stays high while validation accuracy falls.



training fit keeps rising after validation has peaked

One changed patient can rebuild a subtree

Version A

The best root split is oxygen < 92.

Version B

After one borderline case changes, breathing rate becomes the best root split.

Interpretation risk

A path can look precise even when the tree structure is unstable.

Stopping rules limit fragmentation

- Maximum depth limits path length.
- Minimum samples for a split limits tiny parent nodes.
- Minimum samples per leaf limits fragile terminal groups.
- Minimum impurity decrease rejects weak splits.

Tune together

Different rules can produce similar sized trees for different reasons.

Minimum leaf size protects probability estimates

Leaf counts	Estimated positive rate	Reliability
2 of 2	1.00	very uncertain
40 of 50	0.80	more stable
400 of 500	0.80	still population dependent

Same purity, different evidence

Count the observations behind every leaf probability.

Class weights change the split objective

Rare event

Only 2 percent of transactions are fraudulent.

Weighted impurity

Give costly or rare classes more influence when evaluating splits.

Meaning

Weights encode an objective choice. They do not create new positive examples.

A shallow tree in Python

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```
from sklearn.tree import DecisionTreeClassifier

tree = DecisionTreeClassifier(
    criterion="gini",
    max_depth=3,
    min_samples_leaf=20,
    random_state=7,
)
tree.fit(X_train, y_train)
print(tree.get_depth(), tree.get_n_leaves())
print(tree.score(X_train, y_train))
print(tree.score(X_valid, y_valid))
```

Read together

Depth, leaf count, training score, and validation score describe different parts of the fit.

Text rules expose the fitted structure

```
from sklearn.tree import export_text

rules = export_text(tree, feature_names=list(feature_names))
print(rules)
```


Inspection question

Do the root feature and thresholds make sense in the measurement units?

Tree tracing checkpoint

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Applicant

income=8, debt ratio=0.62, prior defaults=0

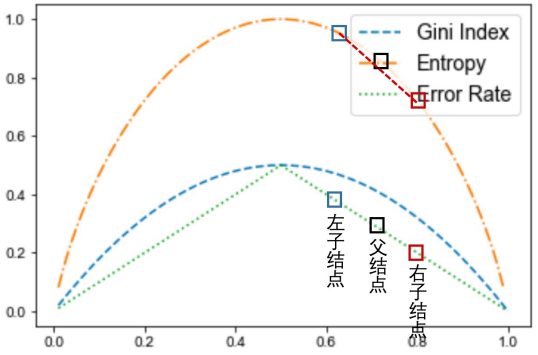
1. Follow the root test.
2. Write every condition on the selected path.
3. Report the leaf sample count and class ratio.
4. Name one nearby change that reaches a different leaf.



Complete answer

A class label without a path and leaf evidence is incomplete.

Gini and entropy measure node impurity



Use weighted child impurity to compare splits; error rate alone can miss a useful improvement.

Entropy measures label uncertainty

$$H(R) = -\sum_{k=1}^{K} p_k \log_2 p_k$$

Pure node

$H = 0$ when one class has probability 1.

Balanced binary node

$H = 1$ bit when $p_1 = p_2 = 0.5$.

Gini impurity is a squared probability measure

$$G(R) = 1 - \sum_{k=1}^{K} p_k^2$$

Binary form

$G = 2p(1 - p)$.

Range

A pure node gives 0. A balanced binary node gives 0.5.

Misclassification error is less sensitive

$$E(R) = 1 - \max_k p_k$$

Use

Useful for describing leaf error.

Split growth

It may treat distinct probability mixtures as equally impure.

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Three impurities on one binary node

Positive share	Entropy	Gini	Error
0.50	1.000	0.500	0.500
0.80	0.722	0.320	0.200
1.00	0.000	0.000	0.000

Shared ordering

All three rank these nodes from most mixed to pure.

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Information gain subtracts weighted child entropy

$$\text{Gain} = H(R) - \frac{n_L}{n} H(R_L) - \frac{n_R}{n} H(R_R)$$

Large gain

The split removes much label uncertainty.

Weighting

Child sizes belong in the formula.

Gini decrease follows the same weighting pattern

$$\Delta G = G(R) - \frac{n_L}{n} G(R_L) - \frac{n_R}{n} G(R_R)$$

Tree choice

Select the feature and threshold with the largest decrease among current candidates.

A hand calculation for one split

Node	Positive	Negative	Gini
Parent	4	4	0.500
Left	3	1	0.375
Right	1	3	0.375

$$\Delta G = 0.5 - \frac{4}{8}(0.375) - \frac{4}{8}(0.375) = 0.125$$

A perfect tiny child may give modest gain

Child	Size	Positive share
Left	1	1.00
Right	99	0.49

Visual trap

The pure child looks impressive but represents only one observation.

Decision

Use the weighted impurity decrease and minimum leaf constraints.

Numeric thresholds come from sorted values

Candidate locations

Consider midpoints between adjacent distinct sorted values.

Values

For 2, 3, 7, and 10, candidates include 2.5, 5, and 8.5.

Duplicate values

Do not split identical values into different branches.

Sorted scans reuse cumulative counts

Naive search

Recount every class on both sides for every threshold.

Efficient scan

Move one sorted sample at a time and update left and right class counts.

Result

Every threshold can be scored in one pass after sorting.

Categorical subset search can explode

Nominal feature

A feature with K categories has many possible binary partitions.

Combinatorial cost

Exhaustive subset search becomes expensive when cardinality is high.

Practical responses

Group rare categories, use ordered statistics carefully, or choose a model with native categorical handling.

Classification and regression trees use binary splits

Classification

A classification tree commonly minimizes Gini impurity.

Regression

A regression tree commonly minimizes squared error within children.

Structure

Every internal node has two outgoing branches.

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Regression splits reduce within node squared error

$$E(R) = \sum_{i:x_i \in R} (y_i - \bar{y}_R)^2$$

$$\Delta E = E(R) - E(R_L) - E(R_R)$$

Leaf prediction

Each child predicts its own mean.

Greedy growth chooses one local winner

At each node

Evaluate candidates and keep the largest immediate impurity reduction.

No backtracking

A standard greedy tree does not revisit an early split after growing later nodes.

Tradeoff

Fast construction can miss a better tree whose first step looks weaker.

The globally best tree is computationally hard

Search space

Tree structures, features, thresholds, and stopping points interact.

Practical strategy

Grow greedily, constrain complexity, and validate or prune.

Claim

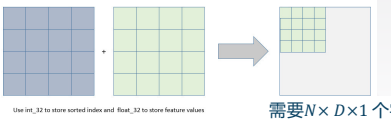
The fitted tree is a useful heuristic solution, not proof of a globally optimal structure.

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Histogram bins reduce the number of candidate splits

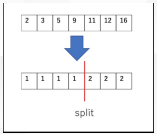
直方图近似确定分割点

直方图可减少存储，且无需排序



需要$N \times D \times 1$ 个字节的内存：仅存储特征离散化后的数值，例如256个bin索引用 8个bit（1个字节）存储

直方图可减少候选分裂点的数目



6个可能的分裂点

2个可能的分裂点

A split search in plain Python

```
def gini(y):
    if len(y) == 0:
        return 0.0
    p = np.bincount(y, minlength=2) / len(y)
    return 1.0 - np.sum(p ** 2)

def split_gain(x, y, threshold):
    left = x <= threshold
    n = len(y)
    child = left.sum()/n * gini(y[left])
    child += (~left).sum()/n * gini(y[~left])
    return gini(y) - child
```

Unit test

A perfect separation should give a larger gain than a random threshold.

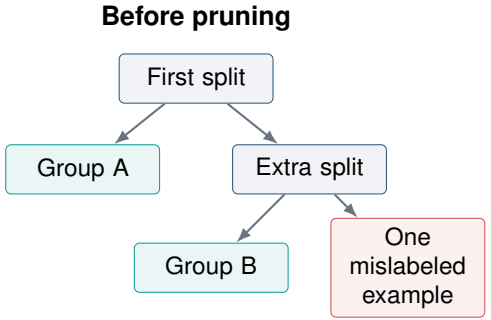
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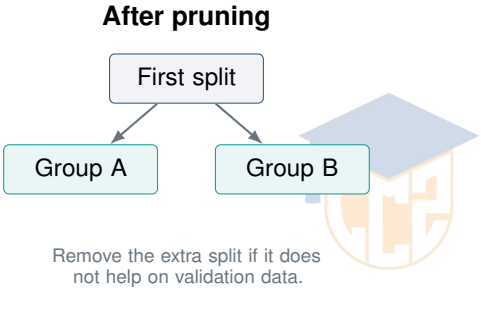
A depth-two tree solves XOR

- First split on $x_1 < 0.5$.
- In each child, split on $x_2 < 0.5$.
- Assign class 1 to $(0, 1)$ and $(1, 0)$; class 0 to the other leaves.
- Each first split has zero impurity gain for balanced XOR: a greedy tree needs a stopping rule that allows this split.



A tree can isolate a mislabeled example





A pure training leaf can be based on only one observation. Check whether the extra rule predicts new examples well.

Prepruning stops before weak branches appear

Controls

Depth, leaf size, split size, and minimum impurity decrease.

Advantage

Training remains faster and the tree stays compact.

Risk

An early weak split may enable valuable later structure, so stopping can be premature.

Postpruning removes branches after growth

Stage 1

Grow a relatively rich tree.

Stage 2

Evaluate nested subtrees and remove branches whose benefit does not justify their complexity.

Advantage

Later evidence can judge whether an early split became useful.

Pruning removes branches that do not help

剪枝

- 分类回归树算法容易过拟合，通过剪枝去除部分分支，降低模型复杂度。
- 剪枝：给定一个完全树，自底向上进行剪枝，直到根节点
- 剪枝准则

对所有叶子结点求和 = 对所有样本求和
（一个样本总会落入某个叶子结点）

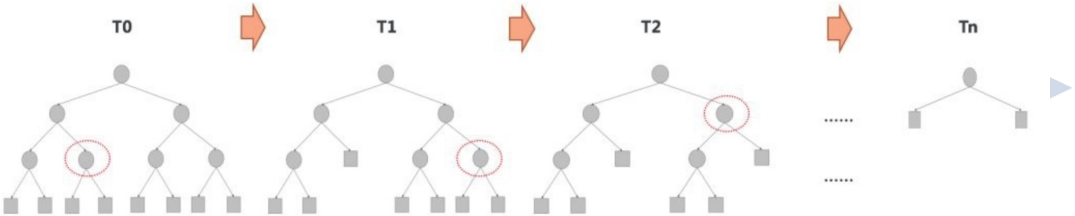
$$C_\alpha(T) = \sum_{t=1}^{|T|} |\mathcal{D}_t| H(\mathcal{D}_t) + \alpha|T| = C(T) + \alpha|T|$$

$|\mathcal{D}_t|H(\mathcal{D}_t)$：叶子结点$t$的不纯净度；$\alpha|T|$：正则因子；$|T|$：叶子结点数目

- 决策树划分得越细致，叶子结点的样本越纯净，但此时叶子结点越多。所以最佳模型是这两方面的折中。
- 当α从0开始增大，树的一些分支被剪掉，得到不同α对应的树

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A pruning path gives smaller subtrees



Pruning gives a sequence of nested subtrees; choose the size using validation data.

Validation selects the subtree

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1. Fit candidate pruning paths on training folds.
2. Evaluate each candidate on held out folds.
3. Choose the complexity rule.
4. Refit on the available training data.
5. Evaluate once on the untouched test set.

Leakage

Pruning on the test set turns the test set into training data.

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Training and validation curves tell different stories

Tree size	Training error	Validation error
small	high	high
moderate	lower	lowest
very large	near zero	higher

Selection

The validation minimum identifies useful complexity, not the training minimum.

Leaf probabilities often need calibration

Raw estimate

Class frequency within a leaf.

Problem

Deep leaves produce extreme 0 or 1 estimates from few samples.

Evaluation

Use calibration curves and proper scoring rules on validation data.

Laplace smoothing tempers tiny leaves

$$\tilde{p}_k = \frac{n_k + \alpha}{n + K\alpha}$$

Binary leaf

For 2 positives out of 2 and $\alpha = 1$, the estimate becomes $3/4$ rather than 1.

Limit

Smoothing helps small counts but does not fix dataset shift.

Impurity importance can favor flexible features

Mechanism

Add impurity decreases from every split that uses a feature.

Bias

Continuous or high cardinality features offer more candidate splits and may appear important by chance.

Check

Compare with validation permutation importance and domain knowledge.

Tree explanations need nearby counterexamples

Path explanation

List the conditions that led to the leaf.

Counterexample

Change one feasible feature and trace whether the route changes.

Feasibility

A mathematically small change may be impossible in the real system.

Small data changes can change a tree

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决策树模型的缺点

- 正确率不高：建树过程过于贪心
  - 可作为Boosting的弱学习器（深度不太深）
- 模型不稳定（方差大）：输入数据小的变化会带来树结构的变化
  - Bagging & 随机森林
- 当特征数目相对样本数目太多时，容易过拟合

Perceptron and tree assumptions

Question	Perceptron	Decision tree
Boundary	one hyperplane	rectangular regions
Learning signal	misclassified samples	impurity decrease
Main risk	nonseparability	high variance
Typical control	representation and stopping	depth and pruning

Choice

Match the model language to the pattern and the cost of mistakes.

Compare a perceptron and a tree

- A perceptron fits one linear score and updates it on mistakes.
- A tree learns feature thresholds and follows a path to a leaf.
- Use held-out examples to compare their errors.

Coding comparison on the same split

```
from sklearn.linear_model import Perceptron
from sklearn.tree import DecisionTreeClassifier

models = {
    "perceptron": Perceptron(max_iter=1000, random_state=7),
    "tree": DecisionTreeClassifier(max_depth=3, min_samples_leaf=10,
                                   random_state=7),
}
for name, model in models.items():
    model.fit(X_train, y_train)
    print(name, model.score(X_train, y_train),
          model.score(X_valid, y_valid))
```

Next step

Inspect errors and structure after the score comparison.

Integrated case for an inspection gate

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Data

Temperature, vibration, operating mode, and 300 labeled inspections.

1. Build a scaled perceptron baseline.
2. Build a shallow pruned tree.
3. Compare costly missed faults.
4. Inspect stability after resampling.



Deployment

Record how missing sensors route through the system.

Failure diagnosis by symptom

Symptom	Likely cause	First check
Perceptron cycles	not separable or bad labels	plot and audit labels
Tree trains perfectly	excess complexity	validation gap and leaf sizes
Tree root keeps changing	correlated features or small data	bootstrap stability
High score disappears	leakage or shift	pipeline and time split

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Check your understanding

- Calculate one perceptron update, including the bias.
- Explain why XOR cannot be separated by one line.
- Choose a tree split using weighted child impurity.
- Explain why pruning can improve predictions on new examples.

Lecture 06 begins with the input representation

Today

We learned how two classifiers use a feature vector.

Next lecture

We will decide how raw values become valid, useful, and leakage free features.

Bridge

The same raw timestamp can become an hour, a weekday, a cyclic pair, or a source of leakage.